FIELD OBSERVATION REPORT

Total Lunar Eclipse (Saros 133) and Nocturnal Microclimatic Stability Assessment

Bosque Sagrado Toyopan, Piedades Sur, Costa Rica

Report ID	LTEMS-OR-2026-003
Report type	Field Observation Report — Astronomical Event Marker
Date	3 March 2026
Observer	Randall Santamaria Artavia, Director-Custodio, Huaravito LTEMS
Location	Bosque Sagrado Toyopan, Piedades Sur, San Ramon, Costa Rica
Coordinates	10.104901 N, 84.565024 W | Elevation: 1,250 m a.s.l.
Monitoring node	Toyopan forest core — Sentinel Microclimate Protocol
Astronomical event	Total Lunar Eclipse, Saros Series 133, 3 March 2026 (pre-dawn)
Dataset status	Integrated into LTEMS March 2026 monitoring sequence

1. Astronomical Context

The pre-dawn hours of 3 March 2026 coincided with a total lunar eclipse belonging to Saros series 133. According to public communication from Cientec Costa Rica, the eclipse occurred approximately 6.5 days prior to lunar perigee, resulting in a relatively large apparent lunar disk against the background constellation Leo. The event is part of the current eclipse season, occurring two weeks after the annular solar eclipse of 17 February 2026.

> *Physical clarification: A lunar eclipse produces no measurable alteration to terrestrial radiative forcing. Lunar albedo contributes negligibly to surface energy balance — estimated at less than 0.002 W/m2 under full-moon conditions, several orders of magnitude below solar irradiance. Consequently, the eclipse is recorded here exclusively as a chronological reference marker within the LTEMS dataset, not as a physical forcing variable.*

The eclipse serves as a well-defined temporal anchor point within the March 2026 monitoring sequence, enabling retrospective identification of the pre-eclipse baseline condition with high precision.

2. Microclimatic Observations — Toyopan Forest Core

Temperature readings were recorded at the Toyopan forest core monitoring node (Sentinel Protocol) across a 12-hour window spanning the eclipse totality phase through the afternoon thermal plateau. Measurements reflect conditions within the forest-influenced microenvironment.

Time (local)	Temperature (°C)	Observation / Phase context
05:00 AM	**16.0 °C**	Eclipse totality phase (peak umbral shadow). Stable pre-dawn nocturnal baseline. Consistent with documented

Time (local)	Temperature (°C)	Observation / Phase context
		LTEMS nocturnal equilibrium range for this elevation and season.
07:00 AM	**18.3 °C**	Post-moonset, pre-sunrise transition. +2.3 °C over 2 hours; within expected range for pre-solar warming at this elevation (see note below).
05:00 PM	**18.7 °C**	Daytime thermal plateau. +0.4 °C over 10 hours of daylight, indicating exceptionally strong thermal inertia within the forest canopy environment.

> *Note on the 05:00–07:00 AM transition: The +2.3 °C increment recorded between totality and post-moonset reflects the standard pre-solar warming curve for the Toyopan site at this elevation and season. This gradient is consistent with prior nocturnal-to-dawn transition records in the LTEMS dataset and does not represent an anomalous departure from baseline conditions.*

Summary metrics

Metric	Value / Interpretation
Pre-dawn baseline temperature (05:00 AM)	16.0 °C — stable nocturnal equilibrium
12-hour temperature range (05:00 AM to 05:00 PM)	+2.7 °C total variation across full observational window
Daytime variation (07:00 AM to 05:00 PM)	+0.4 °C over 10 hours of daylight — strong thermal inertia confirmed
Afternoon plateau (05:00 PM)	18.7 °C — consistent with prior LTEMS thermal buffering records
Thermal inertia classification	High — daytime amplitude below 0.5 °C under current seasonal conditions

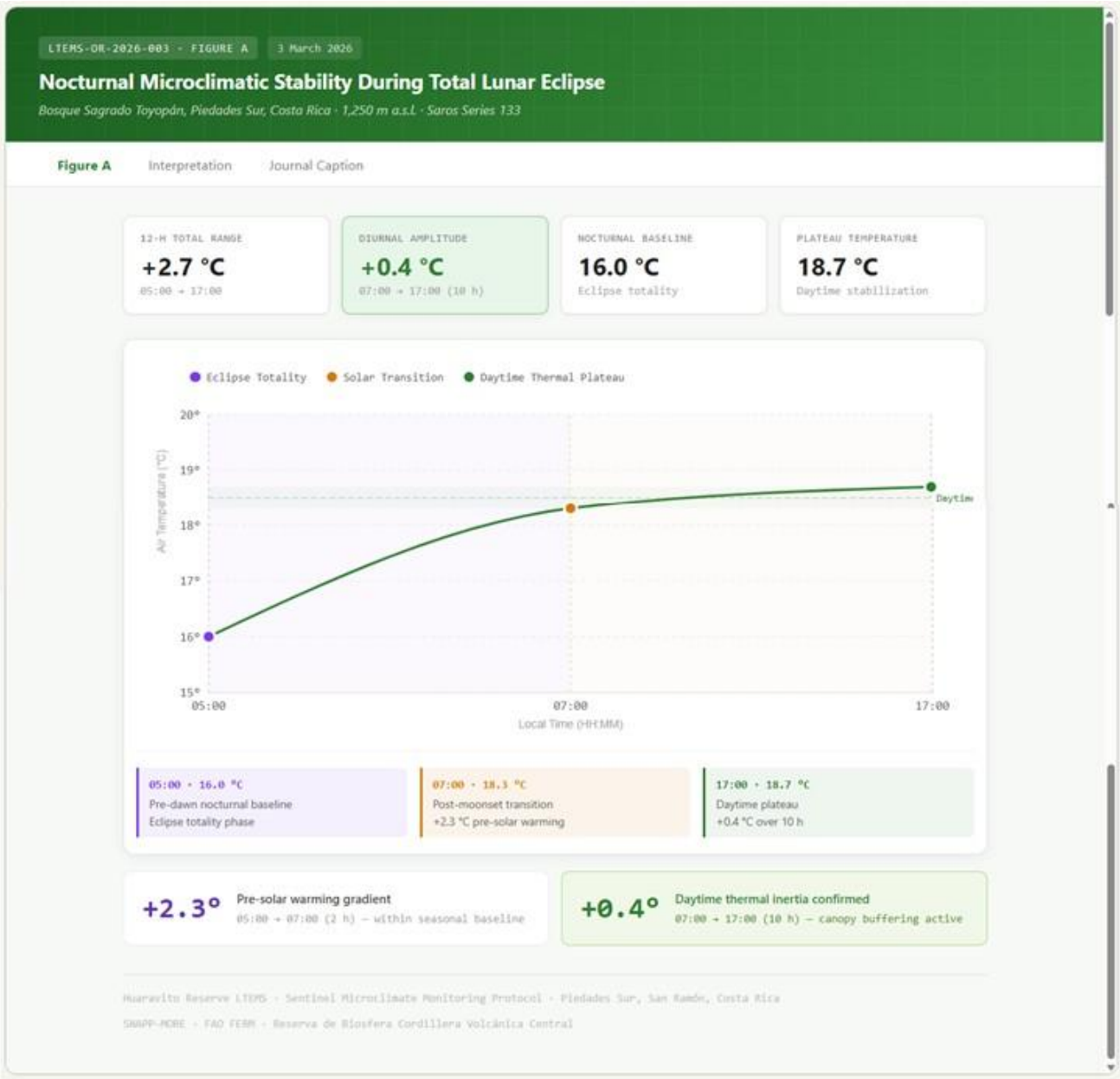
3. Thermal Stability Analysis

No abrupt temperature deviation was detected during the eclipse totality phase or in the subsequent post-moonset transition. The thermal record is continuous and smooth across the full 12-hour observational window, with no inflection points attributable to the astronomical event.

The observed stability reflects three overlapping mechanisms operating independently of lunar illumination:

- Nocturnal radiative cooling equilibrium: the forest canopy moderates longwave emission from the soil surface, maintaining a stable nocturnal temperature regime within the range consistent with prior measurements at this node.
- Canopy thermal buffering: closed-canopy architecture attenuates both incoming and outgoing radiation, reducing the amplitude of diurnal temperature oscillation. The +0.4 °C daytime variation recorded on 3 March 2026 is consistent with this buffering function.
- Soil thermal inertia: subsurface heat storage and release by the forest soil matrix contributes to the damping of short-term temperature fluctuations, particularly under stable atmospheric conditions.

> *Interpretive constraint: The three-point temperature record presented here (05:00 AM, 07:00 AM, 05:00 PM) is sufficient to characterize the general thermal envelope of the observational period but does not constitute a complete diurnal profile. Continuous sensor deployment — as proposed under the Sentinel Protocol expansion — would provide the temporal resolution necessary to resolve sub-hourly dynamics and formally quantify thermal inertia coefficients.*



Online visualization:
https://claude.ai/public/artifacts/c7f4e565-006a-43cf-b2fc-64a1356009b7

4. Atmospheric Transition Context

Regional meteorological forecasts indicate the expected weakening of trade winds and potential intrusion of Saharan dust aerosols within 24 to 48 hours of the observation date. This impending atmospheric transition assigns additional value to the measurements recorded on 3 March 2026, for the following reasons:

- Pre-dust baseline: the measurements establish a clean-sky, low-aerosol-loading temperature profile immediately prior to the anticipated particulate intrusion. This baseline is essential for isolating the thermal signature of Saharan dust if and when it arrives over the Cordillera Volcanica Central.
- Nocturnal clean-sky reference: the eclipse observation provides a well-characterized nocturnal reference point under minimal anthropogenic and natural aerosol interference, which can be compared against post-intrusion nocturnal records.
- Sequence marker: the 3 March 2026 dataset constitutes a formally documented transition boundary within the March monitoring sequence, enabling retrospective comparison once post-dust temperature records are available.

The combination of a precisely timed astronomical event and an impending regional atmospheric shift makes this observation particularly valuable for multi-scale temporal analysis within the LTEMS dataset.

5. Ecological Considerations

No direct physiological thermal stress on vegetation is expected or inferred from the lunar eclipse event. The temperature range recorded across the observational window (16.0–18.7 °C) falls well within the documented thermal tolerance envelope of the premontane forest species assemblage at Toyopan, including the dominant biogenic cooling agent Hauya elegans (Guayabon).

Two indirect ecological dimensions of the eclipse event merit acknowledgment, without implying causal attribution to the observed temperature data:

- Nocturnal fauna behavior: sudden and complete reductions in lunar illumination during eclipse totality are documented in the literature as capable of altering short-term activity patterns in nocturnal fauna (e.g., altered movement, vocalization, or foraging behavior in bats, frogs, and certain invertebrate taxa). This report does not include behavioral observations; such data would require dedicated acoustic monitoring during the eclipse window.
- Photoperception in flora: certain plant species with light-sensitive stomatal responses may exhibit transient adjustments during unusual nocturnal illumination changes. No measurable thermal signature of such responses is expected at the temporal resolution of the present dataset.

> *Caution: No ecological causal relationship between the lunar eclipse and the observed microclimatic data is asserted. The thermal stability documented here is attributable to structural forest properties and seasonal atmospheric conditions, not to the astronomical event.*

6. Sentinel Protocol Relevance

The observations recorded on 3 March 2026 contribute to the Sentinel Microclimate Monitoring Protocol in the following specific ways:

- Nocturnal equilibrium confirmation: the 16.0 °C pre-dawn reading reinforces the documented nocturnal stability profile of the Toyopan forest core, consistent with prior measurements in the LTEMS nocturnal baseline series.

- Diurnal amplitude under current seasonal conditions: the +0.4 °C daytime variation (07:00 AM to 05:00 PM) confirms limited diurnal amplitude under current trade-wind and cloud-cover regimes, consistent with the thermal buffering hypothesis under investigation.
- Afternoon plateau continuity: the 18.7 °C reading at 05:00 PM reinforces prior findings of sustained thermal buffering within the forest nucleus across multiple seasonal windows.
- Pre-transition baseline value: the dataset now formally anchors the pre-dust-intrusion baseline, enabling the Sentinel Protocol to capture a complete before-and-after thermal contrast if Saharan aerosol loading materially alters the regional radiation budget.

This observation exemplifies the value of integrating astronomical and meteorological context into long-term ecological monitoring records: the event itself has no thermal forcing significance, but its precise timing — between a stable baseline period and an impending atmospheric perturbation — maximizes the scientific utility of the measurement.

7. Conclusion

The total lunar eclipse of 3 March 2026 (Saros 133) coincided with stable nocturnal and early-diurnal thermal conditions within the Toyopan forest core. Temperature variation across a 12-hour window remained minimal (+2.7 °C total, with a daytime increment of +0.4 °C over 10 hours), consistent with documented thermal inertia and canopy buffering properties of the Huaravito forest fragment.

The eclipse exerted no detectable influence on the thermal record — as expected from physical principles. Its value to the LTEMS dataset is chronological rather than causal: it provides a precisely timed, formally documented reference point within the March 2026 monitoring sequence, immediately preceding an anticipated regional atmospheric perturbation.

Continued monitoring is recommended to evaluate post-dust-intrusion thermal behavior and to assess whether Saharan aerosol loading produces a detectable signal within the Toyopan microclimate record.

Recommended next monitoring actions

- Continue daily temperature recording at Toyopan node through the anticipated dust-intrusion window (4–6 March 2026).
- Flag any temperature deviation exceeding +1.0 °C from the 3 March 05:00 PM baseline as a potential aerosol-forcing signal.
- If acoustic monitoring equipment is available for future eclipse events, schedule nocturnal biodiversity sampling during totality to document potential fauna behavioral responses.
- Integrate this record into the LTEMS March 2026 monthly synthesis report as a formally documented baseline event.

RESERVA HUARAVITO

Huaravito Reserve LTEMS Field Observation Report

Report ID: LTEMS-OR-2026-003 | Date: 3 March 2026 | Observer: Randall Santamaria Artavia

Sentinel Microclimate Monitoring Protocol | Bosque Sagrado Toyopan, Piedades Sur, San Ramon, Costa Rica

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